

Data Documentation

Session Objectives

By the end of this session, you'll be able to:

- Examine metadata documentation of example projects.
- Explain the purpose of documenting research projects and data.
- Differentiate between project- and data-level documentation.
- List the main contents of README and data dictionary files.
- Explain the difference and give examples of human-readable and machine-readable metadata.

Have you ever gone back to a spreadsheet you've created and can't determine the meaning of your variables or formulas?

Have you found/received data that was confusing or vague?

Share your answers in the chat, or raise a hand!

What is Data Documentation?

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- Data documentation is exactly what it sounds like: a way to document data
- The goal of data documentation is to provide descriptions about data so that it can be correctly interpreted and reused in the future
- This can be both for other people, but also for your future self

What is data documentation?

- Data documentation is exactly what it sounds like: a way to document data
- The goal of data documentation is to provide descriptions about data so that it can be correctly navigated, interpreted, and reused in the future
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Types of Documentation

- There are various ways to document data which are determined by:
 - Disciplinary norms and requirements
 - Standard group or lab practices
 - Types of data (e.g. tabular vs. qualitative data)
 - Internal or external purposes (working with your research group or sharing broadly)

Types of Documentation

- README file
- Data dictionary
- Codebook
- Code file/script
- Standard operating procedures (SOPs)

For the purpose of this course, we're going to focus on the README file

Scenario

A research group that studies geology using computational methods and physical survey data is nearing the end of the academic school year and is close to the summer data-collection season. The group currently has ten graduate students (in different years of study) and two co-principal investigators. Several graduate students will be graduating and leaving the group at the end of the term and a few new grad students will be joining the team. What types of documentation should the research group make sure they have from the students before they leave to ensure that the next cohort of graduate students can be onboarded and continue the research? What details should be included in the documentation?

Share your answer in the chat, or raise a hand!

README Files

What is a README file?

- A plain text file that adds context to a collection of files to help ensure that they can be navigated, interpreted, and understood
- Depending on the nature of the files/project, README files may include slightly different information, but there are some standard fields that cover the basics:

What is a README file?

- Title
- Contact/author
- Description/summary of project
- Software
- File and folder structure
- Naming conventions
- Description of variables
- Methods
- Data confidentiality and permissions
- License and reuse information

How to create a README

README files should:

- Be plain text
- Appear at the top of a file list and be easy to find
 - Tip: Use forced order and prepend filename with an underscore (Ex: **_README.txt**)
- Be updated regularly to ensure accuracy

How to create a README file

README files **should not**:

- Include excessive formatting
- Include irrelevant information
- Include pictures or visual data

Activity: Data Documentation

To better understand how you can document your own project, it is helpful to see how other projects have documented their data. In this activity, you will look at the documentation of different datasets, and discuss some of the characteristics of the documentation as well as any positive or negative aspects that you see.

First, choose one of the following datasets to inspect. You will have ten minutes to read through its documentation and files. Although some people in your break-out group will inspect the same dataset, make sure that each dataset is inspected by at least one person in your group.

Find the links on the webpage: [Activity - explore data documentation](#)

Preparing for Day 2

Preparing for day 2

In the next session, we will use what we have learned so far about data documentation to create a README file for the example project of this course. To prepare for this, make sure to do the following before the start of next day:

- Download the [README file template](#)
- Have the README template open in a text editor so it is ready for the start of the next session.